

Supplementary Note 1: Architectural details of the I2SR and STS modules

I2SR: I2SR is mainly designed for the inter-slice super-resolution for low-resolution CT scans. We utilize 3D convolutional layers with $3 \times 3 \times 1$ kernels and 32 feature maps as the feature extraction (FE) block to obtain the latent representation, and then use the 3D interpolation as the feature interpolation (FI). In our practice, we employ $l=16$ iterations of I2FF blocks, with parameters set to $(k_1, k_2)=(5, 3)$ in our experiments. The detailed architecture is illustrated in Supplementary Figure. 5a. The image reconstruction block consists of two cascaded convolutional layers with $3 \times 3 \times 1$ kernels. A residual connection is established between the middle slice of the input and output results. We employ image augmentation techniques including random flipping, random rotation, random affine transformations, and random histogram shifting during the I2SR training phase with MONAI^[1]. The Adam optimizer is used for the segmentation network with an initial learning rate of 1×10^{-4} . The decay of the first-order momentum gradient is 0.9, and the decay of the second-order momentum gradient is 0.999.

We define two super-resolution specifications to convert from 2.00 mm and 5.00 mm to 1.00 mm, respectively. For CT scans with inter-slice thicknesses different from these specifications, we first interpolate them to the nearest specified thickness before applying I2SR. We then perform STS segmentation on the resampled CT scans to obtain artery-vein segmentation results. The predicted segmentations will be resampled back to the original input resolution to compare our segmentation results and the ground truth labeled on raw resolution space.

STS: The architectural details of the segmentation network used in STS are illustrated in Supplementary Figure 5.c and d. Firstly, the input volume is processed with a $3 \times 3 \times 3$ convolution with 32 channels to generate the initial embeddings. The encoder is composed of successive convolutional blocks with filter sizes of 32, 64, 128, and 256, each followed by a $2 \times$ downsampling block that reduces the spatial dimensions by half. Each convolutional block includes an expansion layer and a paired compression layer. The expansion layer employs an overcomplete convolution with RC output channels, where R is the channel expansion ratio, followed by GELU activation. This channel expansion enables the network to improve its expressiveness and nonlinear representation while maintaining $1 \times 1 \times 1$ convolutions to optimize computational resources^[2]. The compression layer then reduces the channels to restore the feature maps. In our implementation, we set $R=2$. The encoder produces a downsampled embedding, followed by a bottleneck convolution layer with 512 filters and a $3 \times 3 \times 3$ kernel. The decoder mirrors the encoder's structure, replacing downsampling blocks with upsampling blocks, using transpose convolutional operators for $2 \times$ upsampling to recover spatial dimensions. Skip connections link corresponding layers in the encoder and decoder to enhance feature propagation. The final output is a segmentation probability map, accompanied by a prior-map transmission to yield the final segmentation result.

The prior-map transmission block consists of two $3 \times 3 \times 3$ convolutional layers with residual connection. It receives the results of the lower-level artery-vein segmentation and concatenates them with the input CT scan. The outputs after the prior-map transmission module will act as the input for the higher-level segmentation network. The training of STS is respectively performed for each vessel level. We employ image augmentation techniques including random flipping, random channel-wise Gaussian sharpening, and Gaussian noise, operating on both CT data and segmentation prior, during the segmentation training phase. The Adam optimizer is used for the segmentation network with an initial learning rate of 1×10^{-4} . The decay of the first-order momentum gradient is 0.9, and the decay of the second-order momentum gradient is 0.999.

Supplementary Note 2: Training and evaluation of super-resolution

In this note, we explore the effectiveness of I2SR in image super-resolution and its impact on improving artery-vein segmentation on low-resolution CT.

Datasets

We utilized both realistic and simulated paired high-resolution (slice thickness ≤ 1.00 mm) and low-resolution (slice thickness ≥ 2.00 mm) CT scans to develop I2SR. For the realistic paired data, we collected datasets from four hospitals. The paired high-resolution and low-resolution CT scans were obtained from one-time scanning using different reconstruction protocols (Supplementary Table 1, $n=1,000$ in total). These datasets were totally different from artery-vein segmentation studies. For the simulated paired data, we downsampled the high-resolution CT scans using trilinear interpolation to generate the low-resolution counterparts. The RAD dataset ($n=13,000$) was employed for this simulation. We established two super-resolution specifications, which reduce slice thickness from 2.00 mm and 5.00 mm to 1.00 mm, respectively. For scans with other slice thicknesses larger than 2.00 mm, we first resampled them to the nearest set specification using trilinear interpolation and then performed I2SR. To evaluate performance, we randomly selected 200 CT scans with 2.00 mm slice thickness and 200 CT scans with 5.00 mm for testing. The remaining scans were used as training and validation data.

Datasets ID	Data source	Number	Slice thickness (mm)	Vendor
1	The First Affiliated Hospital of Harbin Medical University	151	1.00&2.00&5.00	SIEMENS
2	The Fourth Affiliated Hospital of Harbin Medical University	192	[0.625, 1.00]&2.00&5.00	SIEMENS
3	Mudanjiang First People's Hospital	157	1.00&1.25&5.00	Philips
4	Shanghai Renji Hospital	500	1.00&1.25&5.00	TOSHIBA

Supplementary Table 1: The realistic paired datasets for the development of I2SR.

Super-resolution evaluation

To assess the image quality of super-resolution, we employ three widely used metrics: peak signal-to-noise ratio (PSNR), structural similarity index measure (SSIM)^[3], and root mean square error (RMSE). The RMSE and PSNR can be calculated using the following formulations.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n [y(i) - \hat{y}(i)]^2$$

$$\text{RMSE} = \sqrt{\text{MSE}}$$

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{1}{\text{MSE}} \right)$$

Here y and $\hat{y}$ are the normalized super-resolved image and reference image, respectively, while i represents the pixels in the image, and n is the total number of all pixels. A higher PSNR and lower RMSE usually indicate a higher similarity between the super-resolved and reference images.

SSIM provides a more perceptually relevant measure between the perceived changes in structural information and contrast, rather than just the absolute errors.

$$\left\{ \begin{array}{l} \text{SSIM} = \frac{(2\mu_y\mu_{\hat{y}} + k_1^2)(2\sigma_{y\hat{y}} + k_2^2)}{(\mu_y^2 + \mu_{\hat{y}}^2 + k_1^2)(\sigma_y^2 + \sigma_{\hat{y}}^2 + k_2^2)} \\ \mu_y = \frac{1}{n} \sum_{i=1}^n y(i) \\ \mu_{\hat{y}} = \frac{1}{n} \sum_{i=1}^n \hat{y}(i) \\ \sigma_y^2 = \frac{1}{n-1} \sum_{i=1}^n (y(i) - \mu_y)^2 \\ \sigma_{\hat{y}}^2 = \frac{1}{n-1} \sum_{i=1}^n (\hat{y}(i) - \mu_{\hat{y}})^2 \end{array} \right.$$

We adopt $(k_1 = 0.01, k_2 = 0.01)$ in our computation. The range of the SSIM is $[0, 1]$, and a higher SSIM always means better structural similarity.

Super-resolution results

We compared I2SR with other SOTA super-resolution methods, including trilinear interpolation, SRResNet^[4], and Cycle-GAN^[5]. Each compared model was implemented based on its respective paper and trained and tested on our datasets. We also examined different iteration numbers l for the I2FF blocks, specifically iterations of 6, 12, and 16. The comparison included both the testing for 2.00 mm and 5.00 mm upsampling. Quantitative comparisons, including PSNR, SSIM, and RMSE, are presented in Supplementary Table 2, while visualization results are shown in Supplementary Figure 1. The results indicate that I2SR achieves the highest scores across all evaluation metrics. Notably, compared to other methods, images upsampled by I2SR show superior vessel consistency with the ground truth, whereas other methods result in significant discrepancies. These findings suggest that I2SR is not only an excellent method for CT resampling to normalized resolution but also an optimal preprocessing step for artery-vein segmentation. In the following experiments, we set the iteration to $l=16$, as this configuration typically yields superior results compared to others.

		2.00 mm upsampling			5.00 mm upsampling		
Model		PSNR	SSIM	RMSE ($\times 0.1$)	PSNR	SSIM	RMSE ($\times 0.1$)
Trilinear interpolation		37.79 $\pm$ 5.91	0.967 $\pm$ 0.020	0.156 $\pm$ 0.087	29.76 $\pm$ 4.52	0.892 $\pm$ 0.058	0.363 $\pm$ 0.149
SRResNet		39.65 $\pm$ 4.31	0.977 $\pm$ 0.016	0.112 $\pm$ 0.032	33.74 $\pm$ 6.88	0.921 $\pm$ 0.018	0.219 $\pm$ 0.101
Cycle-GAN		41.02 $\pm$ 1.68	0.983 $\pm$ 0.005	0.097 $\pm$ 0.023	36.32 $\pm$ 2.01	0.955 $\pm$ 0.012	0.179 $\pm$ 0.064
I2SR	Iteration=6	41.87 $\pm$ 1.50	0.988 $\pm$ 0.002	0.082 $\pm$ 0.015	35.89 $\pm$ 1.98	0.952 $\pm$ 0.011	0.167 $\pm$ 0.048
	Iteration=12	42.25 $\pm$ 1.51	0.989$\pm$0.002	0.078$\pm$0.016	37.77 $\pm$ 1.73	0.961 $\pm$ 0.010	0.153 $\pm$ 0.026
	Iteration=16	42.65$\pm$1.31	0.988 $\pm$ 0.002	0.079 $\pm$ 0.018	38.17$\pm$1.71	0.965$\pm$0.008	0.136$\pm$0.029

Supplementary Table 2: Quantitative results for super-resolution. We show the results in mean $\pm$ std. The best performances are highlighted in bold.

Ablation study for artery-vein segmentation

In this section, we present two experiments on the STS artery-vein segmentation applied to resampled CT volumes with different super-resolution methods, to evaluate the necessity of resampling on segmentation performances. First, we evaluate performance on synthesized low-resolution CT data by downsampling normal-resolution CT (inter-slice thickness = 1.00 mm) to low-resolution CT (inter-slice thickness = 5.00 mm). These datasets are then upsampled back to the normalized resolution using different super-resolution methods. The STS segmentation module is applied to these datasets for artery-vein segmentation, and the results are compared against the ground truth. This experiment utilizes all 142 normal-resolution CTs in the test set.

Super-resolution method	Synthesized 5.00 mm CT (n=142)				Low-resolution 2.00 mm CT (n=20)			
	DSC (%)							
	Whole PA	Whole PV	Intra PA	Intra PV	Whole PA	Whole PV	Intra PA	Intra PV
Baseline (without upsampling)	-	-	-	-	58.61±7.95	54.79±7.31	45.05±6.68	50.24±7.50
Trilinear interpolation	67.11±6.02	66.96±6.72	48.98±4.12	51.21±5.02	86.98±5.44	85.55±5.53	78.76±4.38	79.09±5.01
SRResNet	77.69±5.16	77.43±5.75	63.34±4.25	65.40±3.75	88.71±4.41	85.84±5.61	80.92±2.96	81.26±5.66
Cycle-GAN	83.30±3.14	82.61±3.47	72.99±3.35	74.75±3.03	89.25±4.80	87.25±4.94	80.79±2.63	82.05±5.38
I2SR	88.98±1.99	87.84±2.07	79.46±2.75	80.26±2.50	89.15±4.70	87.65±5.49	82.62±2.93	82.97±5.73

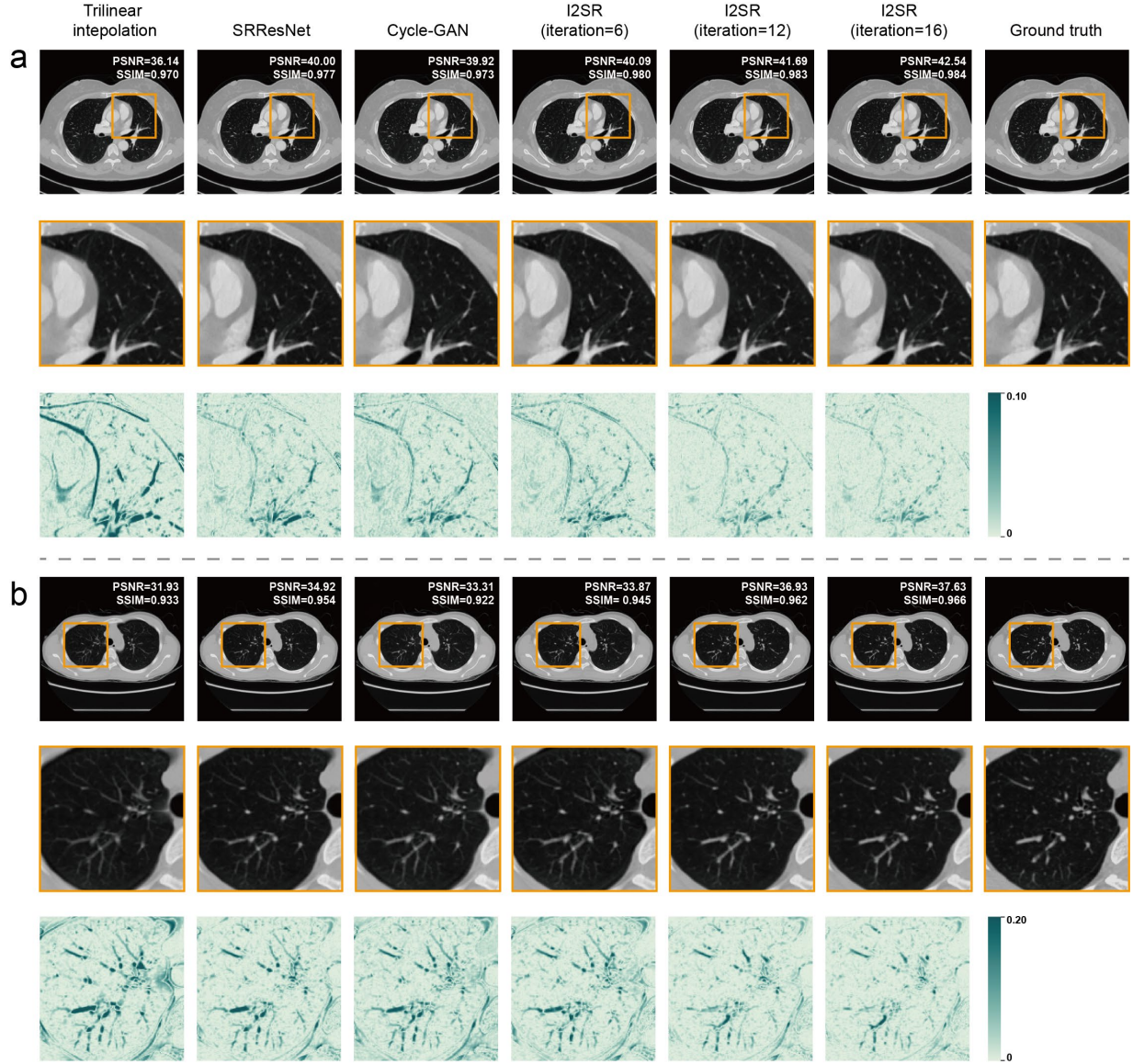
Supplementary Table 3: Segmentation performance of CT resampled with different super-resolution methods.

We compare the performance of artery-vein segmentation using STS on both synthesized 5.00 mm CT and low-resolution 2.00 mm CT. The CT data are resampled to the normalized resolution using various super-resolution methods. For the ablation study on spatial normalization, we include a baseline performance on the low-resolution 2.00 mm CT, achieved by applying STS directly without resampling. Here “PA” = pulmonary artery and “PV” = pulmonary vein. “Intra PA” = intrapulmonary artery, “Intra PV” = intrapulmonary vein, “DSC” = dice similarity coefficient. We show the results in mean $\pm$ std. The best performances are highlighted in bold.

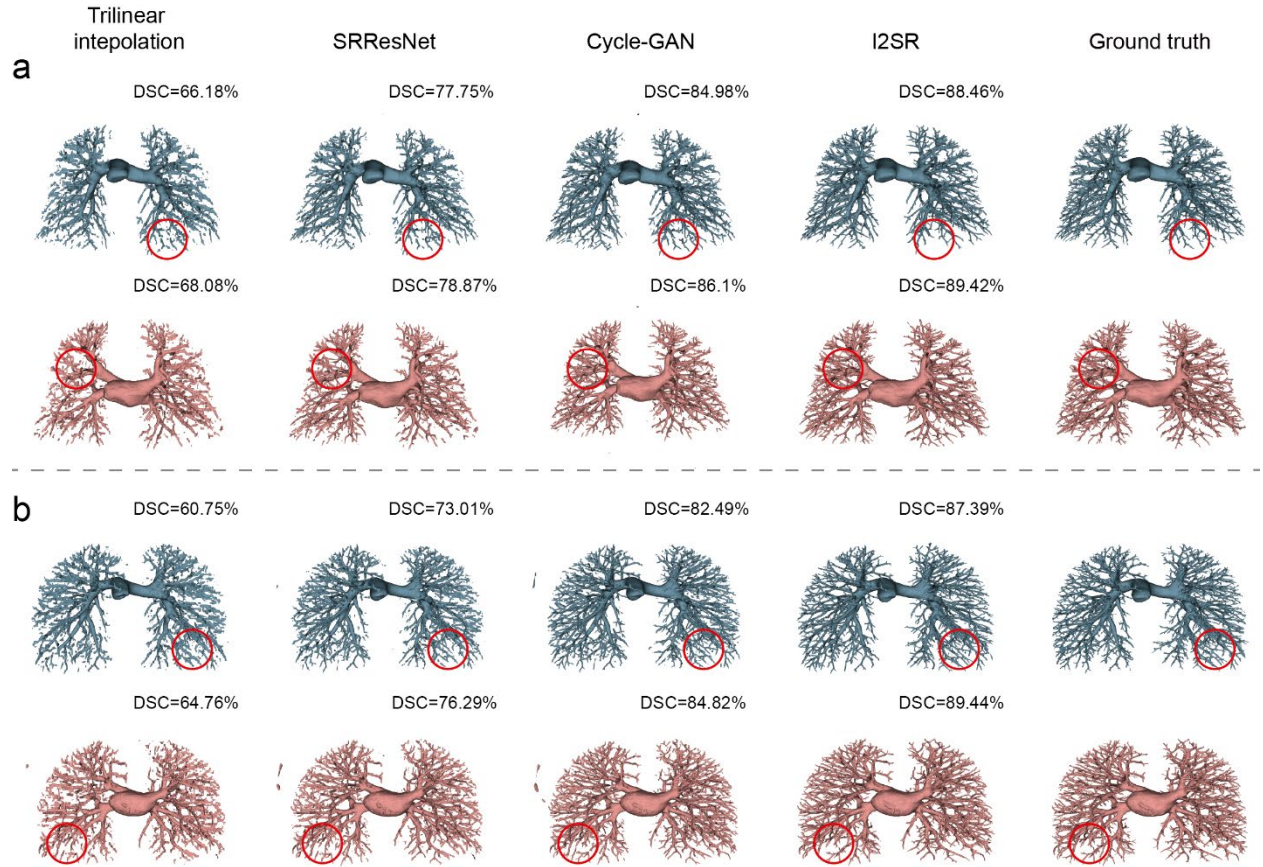
In the second experiment, we compare segmentation performance on low-resolution CT with an inter-slice thickness of 2.00 mm. We first report the baseline by directly applying STS to the raw low-resolution CT. Subsequently, super-resolution methods are applied to the original CT to resample them to normalized resolution, followed by STS

for artery-vein segmentation. These results are then resampled to the original resolution for comparison with the ground truth. We employ 20 CT volumes with 2.00 mm inter-slice thickness for this experiment. We include metrics such as the DSC for whole arteries and veins, as well as for intrapulmonary arteries and veins for comparison. Quantitative comparisons for these two experiments are presented in Supplementary Table 3, with visualization results in Supplementary Figure 3 and 4.

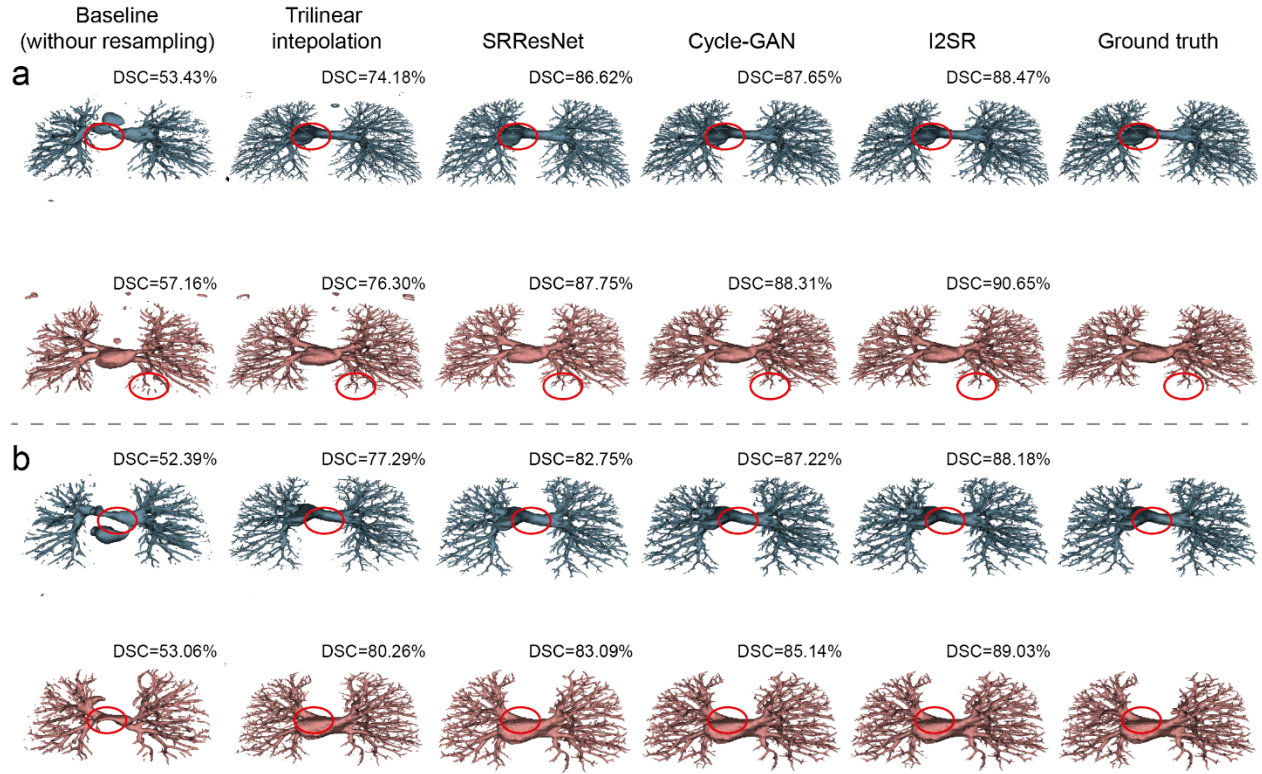
Our experiments demonstrate that applying HiPaS directly to low-resolution CT scans can result in substantial segmentation failure, with the DSC dropping by more than 30% for both arteries and veins. This failure may be primarily due to spatial anisotropy caused by the imbalance in intra- and inter-slice resolution^[6]. Such anisotropy severely degrades the performance of 3D models because the 3D convolutional operator struggles to balance the representation of intra-slice information and discontinuous inter-slice sparse information effectively. Resampling to the normalized resolution markedly improves performance by balancing the intra-slice and inter-slice resolution. Among various super-resolution methods, CT resampled by I2SR achieves the best segmentation outcomes. This is evident not only in quantitative experiments, where I2SR typically attains the highest DSC values for both whole and intrapulmonary regions (Supplementary Table 3), but also in visualization results (see Supplementary Figures 3 and 4). I2SR resampling enables segmentation to maintain a more consistent structure and continuous topology, whereas other methods may produce incorrect or fragmented results. Notably, after I2SR resampling, both 2.00 mm and 5.00 mm CT scans achieve segmentation performance comparable to normal-resolution CT. This indicates that our pipeline, integrating I2SR and STS, can be effectively applied to CT scans across various resolutions, delivering consistently reliable segmentation results.



Supplementary Figure 1: Visualization results for super-resolution. We present the super-resolution results for 2.00 mm upsampling (a) and 5.00 mm upsampling (b), respectively. Figures in the orange box are zoomed in for better visualization. We also show the differences between the super-resolution results and the ground truth below. The PSNR and SSIM values are specific at the right top of each image.



Supplementary Figure 2: Visualization for artery-vein segmentation results on synthesized 5.00 mm thickness CT. We resample the low-resolution CT to normalized resolution with different techniques, and the 3D renderings of segmentation results achieved by HiPaS are presented. (a) and (b) comes from two examples. Arteries are colored with blue, and veins are colored with red. DSC values are specific at the right top of the segmentation result. Red circles mark the noteworthy branches, where I2SR can achieve better segmentation results than other methods.



Supplementary Figure 3: Visualization for artery-vein segmentation results on low-resolution 2.00 mm thickness CT. We resample the low-resolution CT to normalized resolution with different techniques, and the 3D renderings of segmentation results achieved by HiPaS are presented. We also include the segmentation result without any resampling preprocessing as the comparison. (a) and (b) comes from two examples. Arteries are colored with blue, and veins are colored with red. DSC values are specific at the right top of the segmentation result. Results are rescaled into an appropriate ratio for better visualization. Red circles mark the noteworthy branches, where I2SR can achieve better segmentation results than other methods.

Supplementary Note 3: Clinical utility of pulmonary artery-vein segmentation

In this section, we provide a comprehensive introduction to the clinical utility of pulmonary artery-vein segmentation, aiming to help readers appreciate the significance of HiPaS in clinical practice. The clinical utility encompasses three primary aspects: diagnosis and monitoring of pulmonary and cardiovascular diseases, pre-surgical and interventional planning, and acting as the prior for other computer-aided diagnosis (CAD) tasks.

1. Diagnosis and monitoring of pulmonary and cardiovascular diseases

Pulmonary and cardiovascular diseases can severely impact human health, with many such conditions closely associated with the pulmonary arteries and veins. Accurate artery-vein segmentation provides radiologists with the ability to precisely locate potential lesions and assess disease severity by detecting density and morphological changes in arteries and veins. For example, pulmonary embolism, a life-threatening condition, involves blockages that occur exclusively within the pulmonary arteries. By separating arteries from veins, segmentation reduces the search area for these blockages^{[7][8]}, while the detected abnormalities located within the pulmonary veins can also be excluded from being pulmonary embolisms to reduce potential misdiagnosis. Segmentation also facilitates the measurement of vessel morphology, including skeleton length, total branch counts, vessel diameters, and wall thickness^{[9],[10]}. These characteristics are crucial for assessing the severity of pulmonary diseases, such as pulmonary artery hypertension (PAH) and chronic obstructive pulmonary disease (COPD)^{[11][12]}. PAH, for instance, is characterized by the remodeling of small pulmonary arteries, involving the thickening of the intimal and medial layers of the muscular vessel walls^[11]. Recent studies have confirmed that PAH can cause an increase in peripheral pulmonary vessel volume and removal of distal vessels^[11]. Such removal has been shown to be related to the severity of PAH in patients with COPD^[13] and is correlated with worse survival^[14]. Detecting changes in artery morphology can aid in determining disease severity and risk stratification for PAH.

2. Pre-surgical and interventional planning

Artery-vein segmentation is critical in planning complex surgical procedures, including lobectomy, lung tumor resection, and pulmonary embolectomy^[15-17]. It enables surgeons to accurately understand the vascular anatomy and blood supply relationships near lesions, helping to avoid damaging blood vessels during surgery and thereby minimizing the risk of hemorrhage. Notably, both our clinical practice and previous research have reported that females may experience a higher likelihood of hemorrhage and a greater incidence of certain complications during lung surgery or transthoracic needle lung biopsy compared to males^[18]. This disparity can be attributed to anatomical differences between males and females, with females often presenting higher blood vessel density as our study revealed. Precise and detailed artery-vein segmentation can provide a more fine-grained pre-surgical understanding, particularly for female patients, thereby helping to reduce potential surgical risks. Artery-vein segmentation also provides significant benefits for interventional procedures, such as catheter-based treatments for pulmonary embolism^[19], pulmonary angioplasty for pulmonary hypertension^[20], and pulmonary vein isolation for treating atrial fibrillation^[21]. Segmentation serves as a roadmap for interventional radiologists, guiding catheter placement and other interventions with greater precision. This enhanced guidance helps in minimizing procedural risks and improving the success rates of these interventions^[22].

3. Prior to other CAD tasks

Artery-vein segmentation also serves as an important biomarker and prior in numerous CAD tasks and image analysis problems. For instance, the anatomical bifurcations of blood vessels provide crucial biomarkers for inter-patient registration^[23], while comprehensive artery-vein segmentation offers a strong constraint for intra-patient registration^[24]. Additionally, artery segmentation acts as an anatomical prior for identifying and segmenting pulmonary embolisms^[25]. Segmented vessels also enhance anatomical-preserving image restorations, such as super-resolution or denoising^[26]. Moreover, artery-vein segmentation forms the basis for research and analysis of pulmonary hemodynamics^[27]. Collectively, precise segmentation is beneficial for numerous downstream CAD tasks.

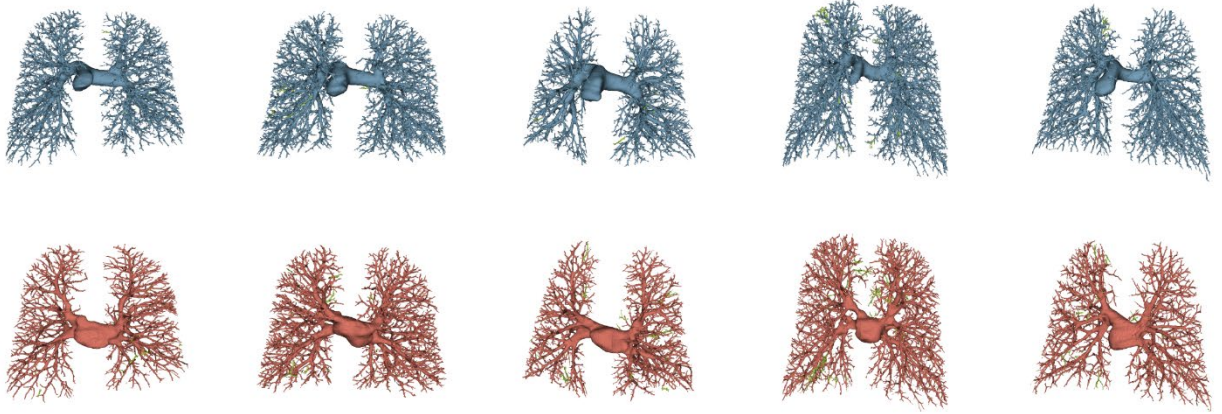
Supplementary Note 4: Comparison of the CARVE14 dataset

The CARVE14 dataset^[28] is a public chest CT dataset comprising 55 non-contrast CT scans. Each scan has an axial size of 512×512 pixels, a slice thickness of 0.7 mm, and spatial resolutions ranging from 0.59 to 0.83 mm. It includes 10 CT scans with full artery-vein annotation and 45 scans with partial annotation of 50 randomly selected vessel subtrees. To demonstrate the robustness of HiPaS on external datasets, we directly deploy it on the CARVE14 dataset without fine-tuning, and then report artery-vein identification performance on both fully and partially annotated cases. Due to differences in annotation standards, where CARVE14 annotations only include intrapulmonary parts, while ours includes the whole pulmonary vessels with generally more branches, we could not compare metrics like segmentation DSC and specificity. Following [28], we use metrics of mean and median classification accuracy (ACC) to evaluate the accuracy of identify artery and vein branches. For example, if artery branches are unidentified or identified as veins, we regard this as a classification error, and vice versa. We also include sensitivity (SEN) for better comparison.

We compare HiPaS with methods by Charbonnier et al.^[28], Nardelli et al.^[29], and Qin et al.^[30], using results directly quoted from their papers. Comparisons are performed for both the 10 fully annotated and 45 partially annotated cases. The quantitative and qualitative results are presented in Supplementary Table 4 and Supplementary Figure 4, respectively.

	10 full-annotation CT			45 partial-annotation CT		
Method	ACC mean (%)	ACC median (%)	SEN (%)	ACC mean (%)	ACC median (%)	SEN (%)
Charbonnier et al.	92 [88, 95]	94 [84, 96]	-	89 [87, 91]	89 [87, 93]	-
Nardelli et al.	94 [91, 96]	95 [93, 97]	97	91 [-, -]	93 [-, -]	95
Qin et al.	90 [88, 93]	-	90	-	-	-
HiPaS	95 [92, 97]	97 [95, 99]	98	92 [91, 93]	94 [91, 95]	98

Supplementary Table 4: The quantitative results for the artery-vein segmentation of the CARVE14 dataset. Here “ACC” = accuracy and “SEN” = sensitivity. We report the ACC along with its 95% Confidence Interval (CI). Data not included in the paper are marked with “-”. The best performances are highlighted in bold.



Supplementary Figure 4: Visualization for artery-vein segmentation results on CARVE14 dataset. We deploy HiPaS directly on the CARVE14 dataset to achieve whole pulmonary artery-vein segmentation. Arteries are marked in blue, veins in red, and unidentified or misclassified intrapulmonary vessels (where arteries are identified as veins, and vice versa) against the full annotations of CARVE14 are marked in green.

The comparison of HiPaS against annotations yields a mean ACC of 94%, a median ACC of 97%, and a SEN of 98% for the 10 fully annotated cases. For the 45 partially annotated cases, HiPaS achieves a mean ACC of 92%, a median ACC of 94%, and a SEN of 98%. HiPaS substantially outperforms other SOTA methods across all evaluation metrics. This is also evident in the visualization results, where only minor errors occur within both arteries and veins. Notably, the methods by Charbonnier et al.^[28] and Nardelli et al.^[29] require pre-segmented blood vessels, a constraint that HiPaS does not have. Overall, HiPaS demonstrates superior performance and greater applicability than other SOTA methods on the CARVE14 dataset.

Supplementary Note 5: Related work

In this section, we present a concise overview of the existing methods for artery-vein segmentation, aiming to inspire further advancements in this field. This background will help readers understand the development of artery-vein segmentation techniques and our contributions to this work.

For accurate artery-vein segmentation in clinical applications, CTPA has first been introduced to enhance the scanning signals and image contrast of blood vessels^[31]. Specifically, iodine-based contrast agents are injected into a peripheral vein, typically in the arm. These agents have a higher capacity to absorb X-rays compared to surrounding tissues, which increases the attenuation of blood for X-rays upon injection. This differential attenuation creates a pronounced contrast between blood vessels and adjacent tissues, thereby improving their visibility. The CT scanning is then performed during the arterial phase when the contrast agents are predominantly located in the pulmonary arteries^[32]. As a result, the pulmonary arteries appear brighter and more distinct compared to the pulmonary veins. This contrast difference is crucial for distinguishing between arteries and veins.

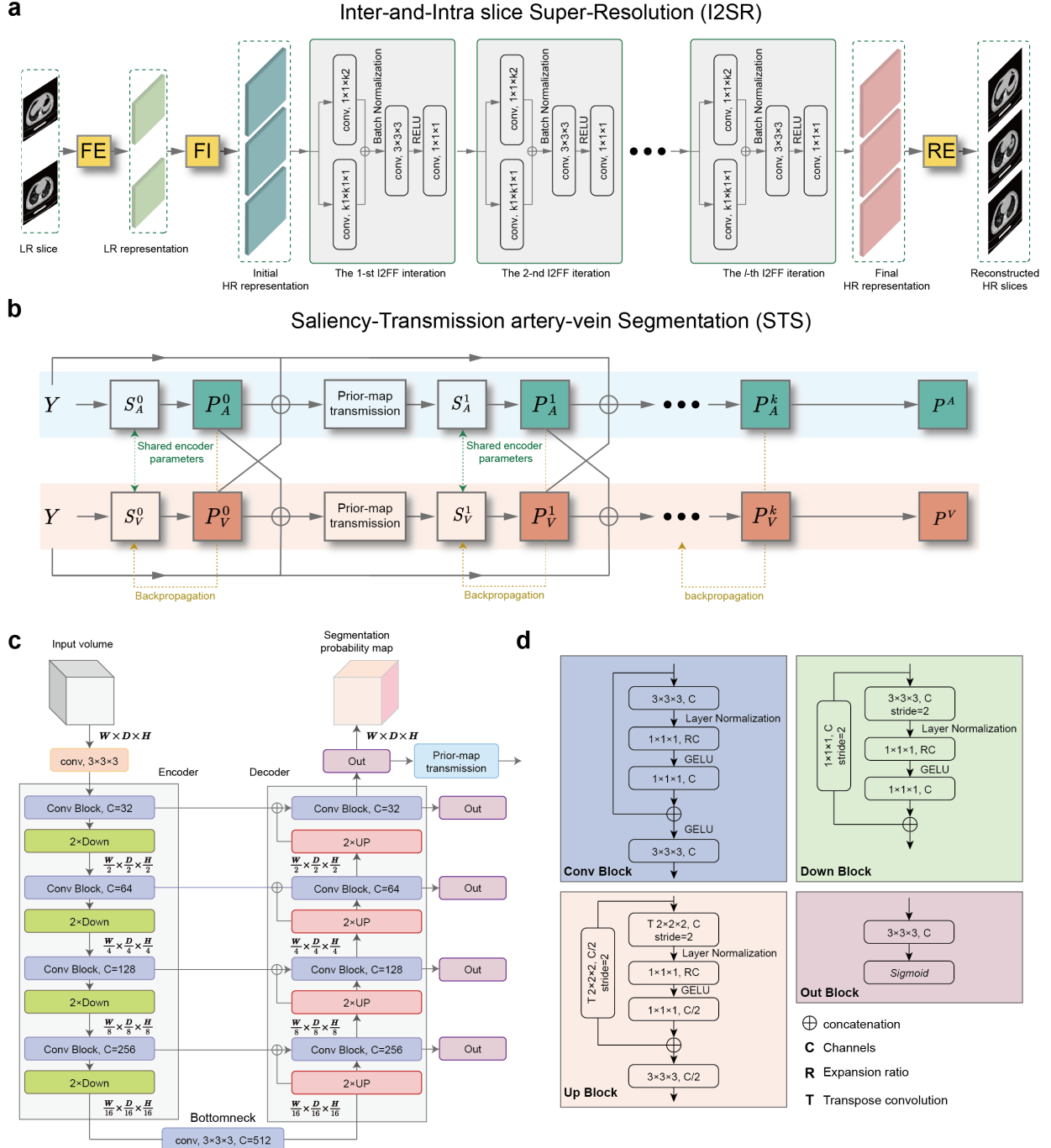
Based on the CTPA technique, some automatic and semi-automatic artery-vein segmentation methods have been introduced. These methods typically exploit the tubular structure of blood vessels and the spatial continuity of arteries and veins to achieve fine-grained segmentation^[33-35]. For instance, Mekada et al.^[33] utilized the anatomical positional relationships between bronchi and blood vessels, along with Voronoi diagrams, to classify arteries and veins. Zhou et al.^[34] developed a 3D multiscale filtering technique and expectation-maximization estimation for pulmonary vessel segmentation and artery-vein classification, incorporating manual tracking of the arterial tree. To address the challenge of misclassification between arteries and veins, Buelow et al.^[35] proposed a seed-point based front-propagation algorithm, which leverages the fact that the pulmonary artery tree accompanies the bronchial tree. However, the use of contrast agents can lead to severe health issues such as allergic reactions, soft-tissue damage, and contrast-induced nephropathy, and are also unsuitable for pregnant women and patients with kidney diseases^{[36][37]}. Consequently, there is an urgent need for a contrast agent-free artery-vein segmentation technique in clinical practice.

With the advancement of artificial intelligence, machine learning and deep learning techniques have made it feasible to segment pulmonary arteries and veins directly from non-contrast CT scans. Charbonnier et al.^[28] pioneered a classification method based on geometric graph representation and subtree separation, which was evaluated compared to expert observers. Subsequently, Nardelli et al.^[29] explored the application of 2D/3D convolutional neural networks (CNNs) and their variants for artery-vein segmentation. However, these methods require prior blood vessel segmentation, which is computationally intensive and time-consuming, while the final performance is highly dependent on the accuracy of vessel extraction, leading to potential error accumulation throughout the processing pipeline. To address this issue, Qin et al.^[30] proposed a tubule-sensitive CNN technique incorporating attention distillation and distance transform maps, enabling direct segmentation on CT scans.

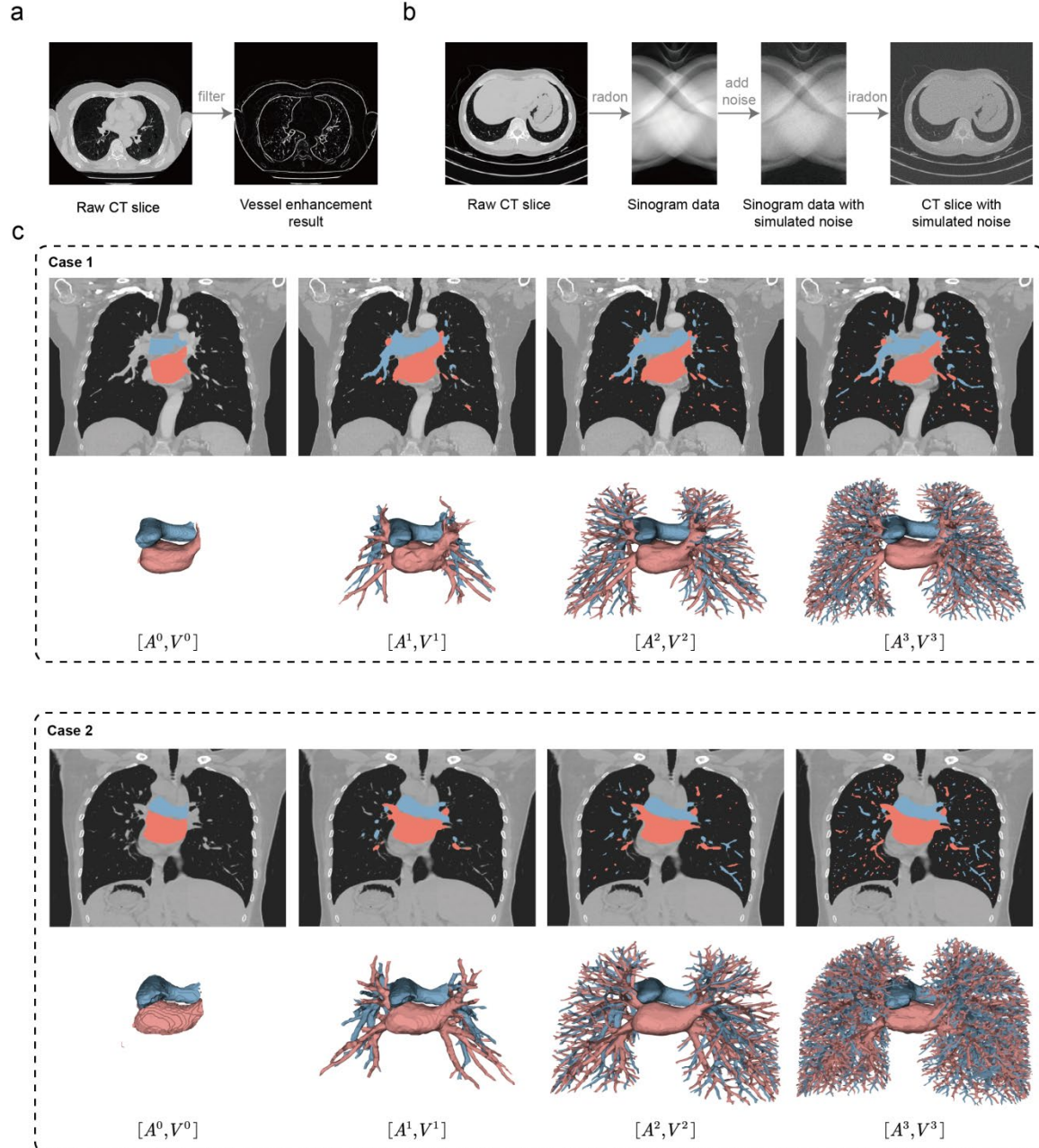
However, the aforementioned methods primarily address intrapulmonary segmentation and cannot process the root arteries and veins at the hilum and regions outside the lung. Identifying these vessels is crucial for diagnosing cardiovascular diseases and pulmonary embolism^{[7][8]}, and it also plays a significant role in planning surgical and interventional procedures, such as coronary artery bypass surgery and heart stent implantation^[39]. Pu et al.^[25] partially addressed this issue by combining a CNN model with differential geometry techniques, successfully segmenting the entire artery and vein trees. To enhance the sensitivity and branch richness of the segmentation, Wang et al.^[24] introduced a radial distance loss, which helps networks focus more on detailed tubular structures. Additionally, Pan et al.^[39] developed a multi-scale information aggregated network, incorporating centerline correction.

Despite advancements, two main challenges still remain. Firstly, current methods often yield less accurate and less abundant segmentation results due to the complex topological structures and low radiographic contrast. The vessel tree exhibits intricate and fine-grained structures in a 3D context^[39]. Existing techniques may result in broken segmentations with artery-vein misclassifications because they typically fail to capture fine-grained tubular structures and long-range spatial correlations^[30]. The low radiographic contrast exacerbates this issue, posing significant challenges for identifying distal vessels, thereby reducing the branch abundance in segmentation results. Diagnosing certain pulmonary diseases, such as pulmonary arterial hypertension or distal lesions, and conducting high-precision surgical planning, relies on accurate full-pulmonary segmentation^{[11][12]}, which current methods cannot provide. Moreover, these methods also struggle to utilize long-range contextual information for distinguishing arteries and veins, which can easily lead to artery-vein misclassification. The second challenge

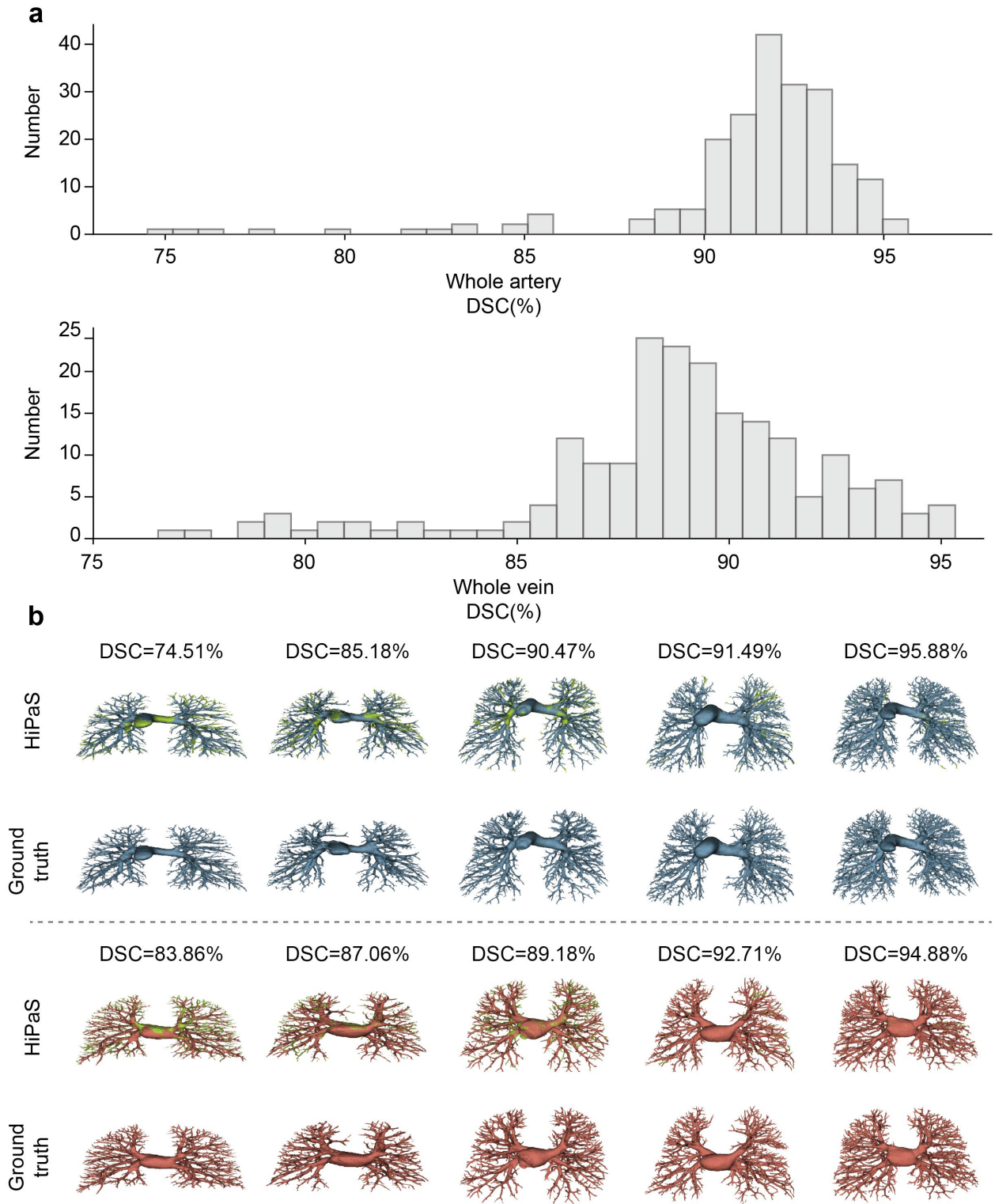
pertains to low-resolution CT scans. Concerns over radiation exposure have led to the increased use of low-dose, low-resolution CT (LD-LRCT). However, low-dose scanning can introduce imaging noise and artifacts that hinder both manual and automated artery-vein segmentation^[40]. Moreover, low-resolution scanning can introduce ambiguity and significant spatial anisotropy, further impairing segmentation performance^[26]. Experiments have shown that this spatial anisotropy negatively affects the final segmentation outcomes^[6]. In our study, we have developed a novel artery-vein segmentation technique, namely HiPaS, which achieves anatomically precise and abundant segmentation results for pulmonary arteries and veins from both normal resolution and LD-LRCT across varied CT protocols.



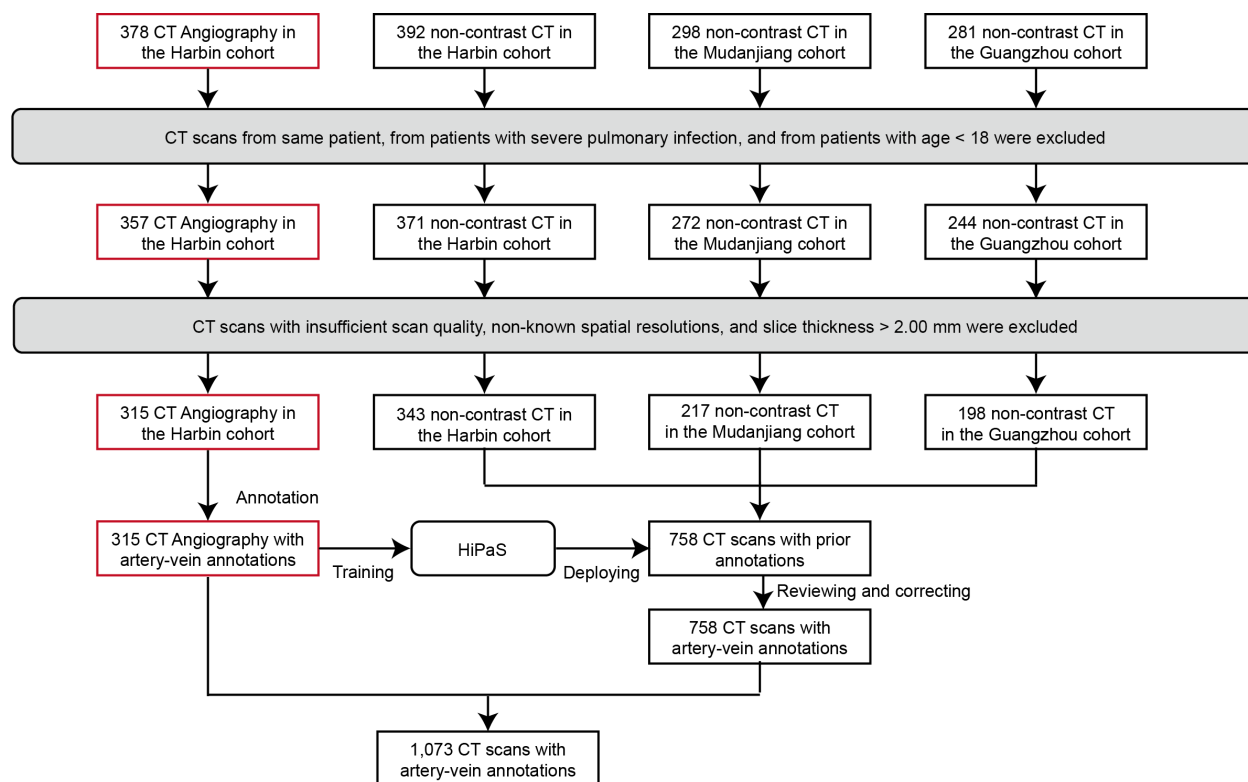
Supplementary Figure 5. Network architecture. **a** The proposed Inter-and-Intra slice Super-Resolution (I2SR) architecture first extracts latent representations of the two input LR slices utilizing a feature extractor (FE) block. The initialized HR representations are then derived through a feature interpolation (FI) block. Inter- and intra-feature fusion (I2FF) modules are subsequently employed to exploit mutual spatial information. The final high-resolution reconstruction (RE) block is employed to reconstruct the target HR slices. **b** Procedure of Saliency-Transmission artery-vein Segmentation. The flow charts with blue and red backgrounds denote the segmentation of arteries and veins, respectively. A cascaded framework progresses from lower-level vessels towards higher-level vessels in a step-wise manner, with probability maps from each stage transmitted as the scaffolds for the next-stage segmentation. The segmentation networks of arteries and veins for each stage have shared parameters of the encoder layers. **c** The detailed structure of the segmentation network S , and **d** the block architectures.



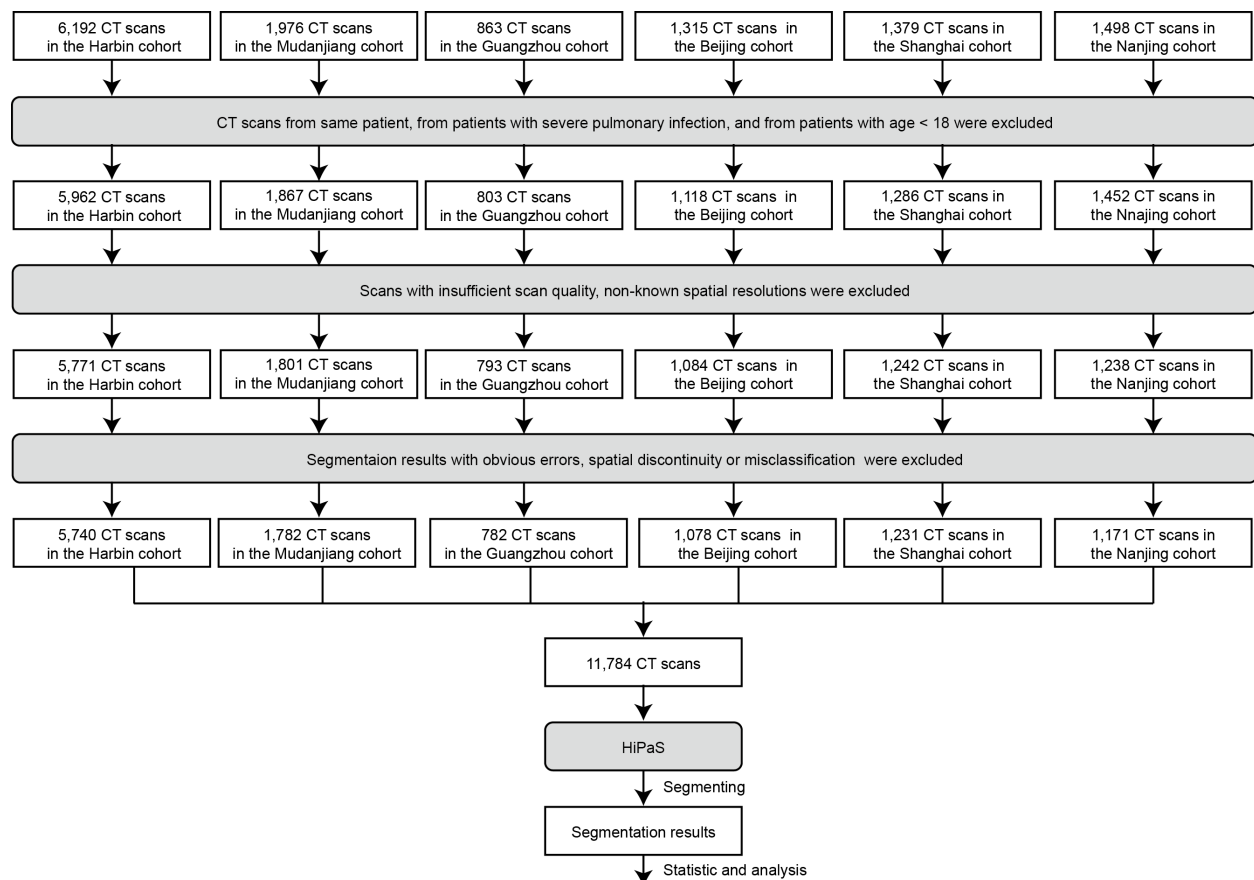
Supplementary Figure 6. Training strategy for HiPaS. **a** An example of the vessel enhancement result after filtering operation. The enhancement result will guide the image reconstruction as well as the artery-vein segmentation. **b** Approach of adding artificial noise onto the CT slices. **c** Two examples of dividing the whole artery-vein trees into four levels. Segmentation for low-level branches will provide a priori to the next-level segmentation.



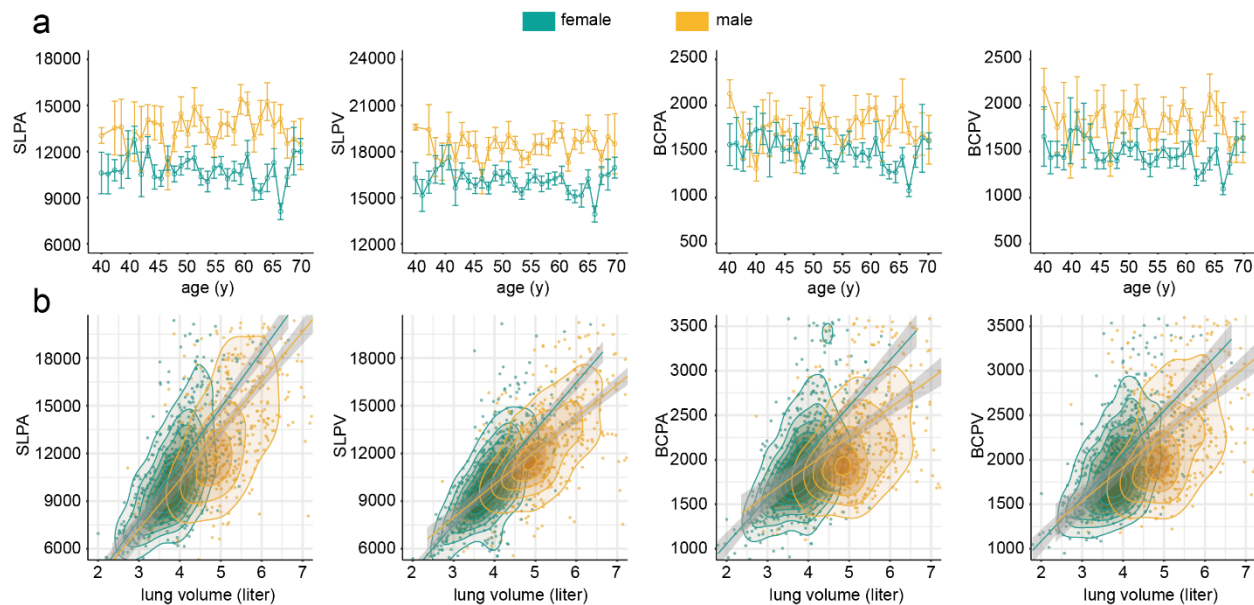
Supplementary Figure 7. More illustrations for artery-vein segmentation performances. (a) Distribution of dice similarity score (DSC) for whole arteries and veins on the test set. Source data are provided as a Source Data file. Source data are provided as a Source Data file. **(b)** The 3D renderings of some representative segmentation results, including the worst, median, and best results. The inconsistency between the ground truth and HiPaS prediction results is marked with green colors. Arteries are presented with blue color, and veins are presented in red. DSC values are specified for each case.



Supplementary Figure 8. Overview of the workflow of the manual artery-vein annotations for 1,073 CT scans.



Supplementary Figure 9. Overview of the workflow of the clinical association cohort for anatomical study.



Supplementary Figure 10. Association of vessel abundance with sex and age on 1,073 CT volumes with manual annotations as ground truth. We include four indices, skeleton length of pulmonary artery (SLPA), skeleton length of pulmonary vein (SLPV), branch count of pulmonary artery (BCPA), and branch count of pulmonary vein (BCPV), and we use the lung volume as the controlling. **a** Pulmonary vessel abundance across different ages. Error bars show the standard error of the mean (SEM). **b** Association and linear regression of vessel abundance compartments with lung volume.

			3DUNet	nnUNet	Pan et al.	Qin et al.	Semi-auto	HiPaS
NRCT (n=142)	DCS(%) ↑	Whole PA	76.22±5.14 ****	80.74±5.06 ****	77.11±4.43 ****	84.85±6.02 ****	85.38±4.99 ****	92.25±1.50
		Whole PV	68.26±7.58 ****	74.95±6.04 ****	72.73±5.17 ****	82.68±7.79 ****	83.51±4.46 ****	89.09±2.10
		Branch PA	62.63±7.78 ****	79.93±8.87 ****	77.91±6.99 ****	78.78±9.70 ****	72.29±8.82 ****	83.50±2.85
		Branch PV	70.15±8.92 ****	80.31±8.79 ****	76.71±8.93 ****	78.50±6.87 ****	77.95±8.69 ****	84.54±2.52
	SEN(%) ↑		77.80±4.95 ****	84.52±4.12 ****	79.46±5.11 ****	87.60±2.18 ****	81.50±4.37 ****	98.77±1.49
	1 - SPE(%) ↓		0.817±0.202 ****	0.571±0.192 ****	0.098±0.051 ****	0.029±0.017 ****	0.175±0.052 ****	0.041±0.018
	MCS(%) ↓		1.88±0.63 ****	2.34±0.85 ****	2.11±0.93 ****	2.05±0.38 ****	1.38±0.33 ****	0.44±0.22
	BN(*100) ↑	PA	2.75±0.61 ****	7.96±2.98 ****	5.52±1.14 ****	6.21±2.70 ****	4.54±2.30 ****	18.57±8.28
		PV	2.65±0.67 ****	6.88±2.97 ****	5.39±1.11 ****	8.79±3.44 ****	6.58±2.91 ****	16.44±6.53
	SL(*1000) ↑	PA	6.03±1.38 ****	11.27±4.83 ****	7.10±2.72 ****	11.51±6.66 ****	9.49±3.28 ****	18.05±6.57
		PV	7.55±1.36 ****	10.21±3.66 ****	8.12±2.60 ****	12.07±4.03 ****	12.56±4.38 ****	15.50±5.79
	HD95 ↓		57.02±47.72 ****	31.65±57.21 ****	37.70±42.53 ****	18.24±16.63 ****	16.06±27.14 ****	4.27±1.17
LRCT (n=56)	DCS(%) ↑	Whole PA	64.10±7.82 ****	68.30±6.21 ****	72.68±8.71 ****	79.08±7.64 ****	76.58±7.56 ****	89.51±5.31
		Whole PV	63.74±4.81 ****	67.40±7.25 ****	71.86±7.20 ****	78.14±7.05 ****	75.23±9.11 ****	88.34±6.28
		Branch PA	57.08±9.09 ****	61.70±7.19 ****	62.16±6.32 ****	67.82±8.34 ****	62.11±8.37 ****	82.56±4.40
		Branch PV	56.85±7.82 ****	62.44±8.22 ****	64.14±9.32 ****	69.28±7.43 ****	63.62±8.46 ****	82.78±6.00
	SEN(%) ↑		57.46±7.51 ****	59.23±7.55 ****	62.51±8.56 ****	78.28±8.60 ****	76.22±6.62 ****	97.24±1.07
	1 - SPE(%) ↓		0.87±0.37 ****	0.432±0.266 ****	0.163±0.08 ****	0.037±0.024 ****	0.084±0.037 ****	0.033±0.019
	MCS(%) ↓		2.17±0.56 ****	1.85±0.86 ****	2.57±0.83 ****	1.24±1.07 ****	1.45±0.81 ****	0.55±0.24
	BN(*100) ↑	PA	2.53±1.03 ****	7.01±2.60 ****	5.87±1.63 ****	6.56±2.76 ****	5.67±2.65 ****	18.20±10.67
		PV	2.33±0.52 ****	6.65±2.54 ****	5.32±1.43 ****	7.54±2.55 ****	6.35±3.07 ****	15.54±7.71
	SL(*1000) ↑	PA	6.08±1.31 ****	10.07±5.38 ****	8.41±2.60 ****	12.87±6.00 ****	10.74±6.74 ****	18.26±10.52
		PV	6.86±1.41 ****	11.34±4.68 ****	8.42±3.61 ****	11.52±4.64 ****	11.10±5.31 ****	16.25±7.21
	HD95 ↓		76.02±78.11 ****	47.70±27.85 ****	42.82±41.31 ****	21.83±22.35 ****	19.63±12.84 ****	4.78±1.89

Supplementary Table 5. Comparison of pulmonary artery-vein segmentation results on normal-resolution CT scans and low-resolution CT scans. Here “NRCT” = normal-resolution CT, “LRCT” = low-resolution CT, “PA” = pulmonary artery, and “PV” = pulmonary vein. “Intra PA” = intrapulmonary artery, “Intra PV” = intrapulmonary vein, “DSC” = dice similarity coefficient, “SEN” = sensitivity, “SPE” = specificity, “MCS” = misclassification score, “BC” = branch counts, “SL” = skeleton length, “HD95” = 95% Hausdorff Distance. The best performances are marked in bold. Here we re-sampled all segmentation results to a standardized space. Two-sided Wilcoxon Signed Ranked tests are done between other methods and our proposed methods. *P* values are specified as: ****p*<0.001, *****p*<0.0001, ns, not significant.

Patient	Diseases	Age	Sex	CT method	Clinical Scenario
1	Lung tumor	53	Male	non-contrast w/o CTPA	inpatient
2	Pulmonary embolism	51	Female	non-contrast w/ CTPA	inpatient
3	Lung tumor	63	Male	non-contrast w/ CTPA	inpatient
4	Lung tumor	57	Female	non-contrast w/ CTPA	inpatient
5	Pulmonary embolism	38	Female	non-contrast w/ CTPA	inpatient
6	Pulmonary artery hypertension	62	Male	non-contrast w/o CTPA	outpatient
7	Pulmonary embolism	55	Female	non-contrast w/ CTPA	inpatient
8	Pulmonary embolism	58	Male	CTPA	inpatient
9	Pulmonary artery hypertension	60	Male	non-contrast w/o CTPA	inpatient
10	Lung nodules	53	Male	non-contrast w/ CTPA	inpatient
11	Pulmonary embolism	56	Female	CTPA	inpatient
12	Pulmonary embolism	60	Male	non-contrast w/ CTPA	inpatient
13	Lung tumor	59	Male	non-contrast w/ CTPA	inpatient
14	Lung tumor	75	Male	non-contrast w/ CTPA	inpatient
15	Chronic obstructive pulmonary disease (COPD)	35	Male	non-contrast w/o CTPA	outpatient
16	Chronic obstructive pulmonary disease (COPD)	49	Female	non-contrast w/o CTPA	outpatient
17	Pulmonary embolism	42	Female	CTPA	inpatient
18	Lung tumor	48	Male	non-contrast w/ CTPA	inpatient
19	Chronic obstructive pulmonary disease (COPD)	60	Female	non-contrast w/o CTPA	inpatient
20	Chronic obstructive pulmonary disease (COPD)	51	Male	non-contrast w/o CTPA	inpatient
21	Pulmonary embolism	57	Male	non-contrast w/ CTPA	inpatient
22	Pneumothorax	28	Male	non-contrast w/ CTPA	inpatient
23	Lung nodules	44	Female	non-contrast w/o CTPA	inpatient
24	Pulmonary embolism	53	Male	CTPA	inpatient
25	Chronic obstructive pulmonary disease (COPD)	46	Male	non-contrast w/o CTPA	inpatient
26	Chronic obstructive pulmonary disease (COPD)	50	Female	non-contrast w/o CTPA	inpatient
27	Lung tumor	72	Female	non-contrast w/ CTPA	inpatient
28	Pulmonary embolism	38	Male	non-contrast w/ CTPA	inpatient
29	Lung tumor	58	Male	non-contrast w/ CTPA	inpatient
30	Lung tumor	55	Male	non-contrast w/ CTPA	inpatient
31	Chronic obstructive pulmonary disease (COPD)	49	Male	non-contrast w/o CTPA	inpatient
32	Chronic obstructive pulmonary disease (COPD)	46	Female	non-contrast w/ CTPA	outpatient
33	Chronic obstructive pulmonary disease (COPD)	56	Male	non-contrast w/o CTPA	inpatient
34	Pulmonary artery hypertension	48	Male	non-contrast w/o CTPA	inpatient
35	Lung tumor	59	Female	non-contrast w/ CTPA	inpatient
36	Lung tumor	76	Male	non-contrast w/ CTPA	inpatient
37	Lung tumor	69	Male	non-contrast w/ CTPA	inpatient
38	Chronic thromboembolic pulmonary hypertension	55	Female	non-contrast w/o CTPA	outpatient
39	Lung nodules	59	Female	non-contrast w/o CTPA	inpatient
40	Chronic obstructive pulmonary disease (COPD)	52	Female	non-contrast w/o CTPA	outpatient
41	Lung tumor	64	Male	non-contrast w/ CTPA	inpatient
42	Pulmonary artery hypertension	58	Male	non-contrast w/o CTPA	inpatient
43	Lung nodules	60	Male	non-contrast w/ CTPA	inpatient
44	Lung tumor	55	Female	non-contrast w/ CTPA	inpatient
45	Pulmonary artery hypertension	55	Female	non-contrast w/o CTPA	outpatient
46	Lung nodules	42	Male	non-contrast w/o CTPA	inpatient
47	Pulmonary embolism	43	Female	non-contrast w/ CTPA	inpatient
48	Pulmonary embolism	48	Female	CTPA	inpatient
49	Pulmonary artery hypertension	57	Male	non-contrast w/o CTPA	outpatient
50	Lung tumor	75	Male	non-contrast w/o CTPA	inpatient

Supplementary Table 6. Patient information of 50 study cases for clinical evaluation. “w/o”=without, “w/”=with.

Accuracy and robustness	
0	The segmentation result is not available or deviates substantially from the ground truth.
1	The segmentation exhibits poor reliability and yields inaccurate or inconsistent results for the principal vascular trunk.
2	Notable deficiencies exist in segmentation accuracy and robustness, failing to accurately delineate the anatomies of the pulmonary artery and vein.
3	Though generally accurate, segmentation contains appreciable errors or misclassifications between arteries and veins.
4	Segmentation proves largely accurate and robust overall, with good resulting vessel morphology, but with minor errors in a few regions
5	Segmentation results demonstrate high accuracy, with highly faithful morphology of the segmented vessels and few errors or inconsistencies.
Branch abundance	
0	The segmentation result only contains the vessel trunks.
1	The segmentation result only contains 2 to 4 levels of vessel branches. Here every forked branch is defined as a new branch level.
2	The segmentation result contains 4 to 5 levels of vessel branches.
3	The segmentation result contains 5 to 7 levels of vessel branches.
4	The segmentation result contains most pulmonary branches but with a few remaining undetected.
5	The result accurately identifies and segments nearly all visible branches of the pulmonary arteries and veins on CT, yielding an overall rich vascular tree morphology.
Assistance for diagnosis	
0	The segmentation outcomes offer negligible diagnostic utility, with the algorithm providing minimal meaningful assistance to the clinician.
1	The algorithm affords limited diagnostic aid and retains substantial room for improvement. Segmentation can assist in diagnosis, yet still necessitates manual adjustments and corrections.
2	It may facilitate surgical planning but demonstrates difficulty in aiding disease diagnosis or severity assessment.
3	Segmentation furnishes valuable diagnostic support, and with a few human adjustments, can assist surgical planning and disease diagnosis.
4	Segmentation provides a valuable asset for diagnosis and surgical planning. The clarity of the results enable clinicians to identify and analyze the structures of pulmonary arteries and veins confidently.
5	Segmentation results address most scenarios of clinical utility and can guide surgical planning and diagnosis with few requiring modifications.

Supplementary Table 7. Scoring rules for clinical evaluations.

	SLPA	SLPV	BCPA	BCPV
total (n=11,784)				
sex	-844.84 ± 56.08 ****	-566.74 ± 40.72 ****	-196.55 ± 11.80 ****	-160.58 ± 10.89 ****
age (y)	-25.18 ± 3.06 ****	-18.71 ± 2.22 ****	-5.04 ± 0.64 ****	-7.79 ± 0.59 ****
lung volume (L)	3149.39 ± 33.95 ****	2413.96 ± 24.65 ****	416.05 ± 7.14 ****	412.77 ± 6.59 ****
Cardiomegaly	171.67 ± 58.18 0.003	150.15 ± 42.25 ****	37.65 ± 12.24 0.002	27.66 ± 11.30 0.014
Artery wall calcification	106.22 ± 53.03 0.045	-18.21 ± 38.50 0.636	9.88 ± 11.16 0.376	-4.78 ± 10.30 0.642
PAH	-191.66 ± 96.38 0.047	7.17 ± 69.98 0.918	-42.88 ± 20.28 0.034	-4.46 ± 18.71 0.812
Pulmonary nodule	13.33 ± 40.66 0.743	1.90 ± 29.52 0.949	10.11 ± 8.55 0.237	-9.17 ± 7.89 0.246
Pulmonary embolism	-8.44 ± 73.30 0.908	50.89 ± 53.22 0.339	9.62 ± 15.42 0.533	17.20 ± 14.23 0.227
COPD	-160.79 ± 102.3 0.116	-170.07 ± 74.28 0.022	-35.12 ± 21.52 0.103	-41.95 ± 19.86 0.035
R2	0.572	0.603	0.303	0.349
p-value	****	****	****	****

Supplementary Table 8. Multiple linear regression models on 11,784 CT participants. We present the linear regression coefficients of demographic characteristics including sex, age, lung volume, and disease states. Sex and disease states are binary encoded for regression (male = 1, female = 0; with disease = 1, without disease = 0). “SLPA” = skeleton length of pulmonary artery, “SLPV” = skeleton length of pulmonary vein, “BCPA” = branch count of pulmonary artery, “BCPV” = branch count of pulmonary vein. “PAH” = pulmonary artery hypertension, “COPD” = chronic obstructive pulmonary disease. The *P*-values are derived from multiple linear regression analysis, indicating the statistical significance of each independent variable. *P* values are specified as: ****p*<0.001, *****p*<0.0001, ns, not significant.

	SLPA	SLPV	BCPA	BCPV
total (n=1,073)				
sex	-956.99±254.00 ****	-740.75±186.42 ****	-220.38±53.21 ****	-165.54±49.99 ****
age (y)	-20.79±10.59 0.049	-15.19±7.77 0.051	-4.48±2.22 0.044	-7.09±2.08 ***
lung volume (L)	3128.87±118.92 ****	2483.73±87.28 ****	409.58±24.91 ****	408.25±23.41 ****
R2	0.463	0.503	0.233	0.269
p-value	****	****	****	****

Supplementary Table 9. Multiple linear regression models on 1,073 CT participants with manual annotation.

We present the linear regression coefficients of demographic characteristics including sex, age, and lung volumes. Sex is binary encoded for regression (male = 1, female = 0). Disease states are not included in the regression because their total number is too small and not representative. “SLPA” = skeleton length of pulmonary artery, “SLPV” = skeleton length of pulmonary vein, “BCPA” = branch count of pulmonary artery, “BCPV” = branch count of pulmonary vein. The *P*-values are derived from multiple linear regression analysis, indicating the statistical significance of each independent variable. *P* values are specified as: ****p*<0.001, *****p*<0.0001, ns, not significant.

References

- [1]. Diaz-Pinto A, Alle S, Nath V, et al. Monai label: A framework for ai-assisted interactive labeling of 3d medical images[J]. Medical Image Analysis, 2024, 95: 103207.
- [2]. Roy S, Koehler G, Ulrich C, et al. Mednext: transformer-driven scaling of convnets for medical image segmentation[C]//International Conference on Medical Image Computing and Computer-Assisted Intervention. Cham: Springer Nature Switzerland, 2023: 405-415.
- [3]. Hore A, Ziou D. Image quality metrics: PSNR vs. SSIM[C]//2010 20th international conference on pattern recognition. IEEE, 2010: 2366-2369.
- [4]. Li J, Fang F, Mei K, et al. Multi-scale residual network for image super-resolution[C]//Proceedings of the European conference on computer vision (ECCV). 2018: 517-532.
- [5]. You C, Li G, Zhang Y, et al. CT super-resolution GAN constrained by the identical, residual, and cycle learning ensemble (GAN-CIRCLE)[J]. IEEE transactions on medical imaging, 2019, 39(1): 188-203.
- [6]. Dong Z, He Y, Qi X, et al. MNet: rethinking 2D/3D networks for anisotropic medical image segmentation[C]//Thirty-First International Joint Conference on Artificial Intelligence {IJCAI-22}. International Joint Conferences on Artificial Intelligence Organization, 2022: 870-876.
- [7]. Su R, van der Sluijs P M, Chen Y, et al. CAVE: Cerebral artery–vein segmentation in digital subtraction angiography[J]. Computerized Medical Imaging and Graphics, 2024, 115: 102392.
- [8]. Pu J, Gezer N S, Ren S, et al. Automated detection and segmentation of pulmonary embolisms on computed tomography pulmonary angiography (CTPA) using deep learning but without manual outlining[J]. Medical Image Analysis, 2023, 89: 102882.
- [9]. Abedin M, Tintut Y, Demer L L. Vascular calcification: mechanisms and clinical ramifications[J]. Arteriosclerosis, thrombosis, and vascular biology, 2004, 24(7): 1161-1170.
- [10]. Shum J, DiMartino E S, Goldhammer A, et al. Semiautomatic vessel wall detection and quantification of wall thickness in computed tomography images of human abdominal aortic aneurysms[J]. Medical physics, 2010, 37(2): 638-648.
- [11]. Shahin Y, Alabed S, Alkhanfar D, et al. Quantitative CT evaluation of small pulmonary vessels has functional and prognostic value in pulmonary hypertension[J]. Radiology, 2022, 305(2): 431-440.
- [12]. Hueper K, Vogel-Claussen J, Parikh M A, et al. Pulmonary microvascular blood flow in mild chronic obstructive pulmonary disease and emphysema. The MESA COPD Study[J]. American journal of respiratory and critical care medicine, 2015, 192(5): 570-580.
- [13]. Sakao S, Voelkel N F, Tatsumi K. The vascular bed in COPD: pulmonary hypertension and pulmonary vascular alterations[J]. European respiratory review, 2014, 23(133): 350-355.
- [14]. Alkhanfar D, Shahin Y, Alandejani F, et al. Severe pulmonary hypertension associated with lung disease is characterised by a loss of small pulmonary vessels on quantitative computed tomography[J]. ERJ open research, 2022, 8(2).
- [15]. Kumar R P. Fast blood vessel segmentation for surgical and interventional planning and navigation[J]. 2014.
- [16]. Valverde I, Gomez G, Gonzalez A, et al. Three-dimensional patient-specific cardiac model for surgical planning in Nikaidoh procedure[J]. Cardiology in the Young, 2015, 25(4): 698-704.
- [17]. Haramati L B, Glickstein J S, Issenberg H J, et al. MR imaging and CT of vascular anomalies and connections in patients with congenital heart disease: significance in surgical planning[J]. Radiographics, 2002, 22(2): 337-349.
- [18]. Tai R, Dunne R M, Trotman-Dickenson B, et al. Frequency and severity of pulmonary hemorrhage in patients undergoing percutaneous CT-guided transthoracic lung biopsy: single-institution experience of 1175 cases[J]. Radiology, 2016, 279(1): 287-296.
- [19]. Engelberger R P, Kucher N. Catheter-based reperfusion treatment of pulmonary embolism[J]. Circulation, 2011, 124(19): 2139-2144.
- [20]. Lang I, Meyer B C, Ogo T, et al. Balloon pulmonary angioplasty in chronic thromboembolic pulmonary hypertension[J]. European Respiratory Review, 2017, 26(143).

- [21]. Khan M N, Jaïs P, Cummings J, et al. Pulmonary-vein isolation for atrial fibrillation in patients with heart failure[J]. New England Journal of Medicine, 2008, 359(17): 1778-1785.
- [22]. Glöckler M, Halbfäß J, Koch A, et al. Multimodality 3D - roadmap for cardiovascular interventions in congenital heart disease—A single - center, retrospective analysis of 78 cases[J]. Catheterization and Cardiovascular Interventions, 2013, 82(3): 436-442.
- [23]. Cazoulat, Guillaume, et al. "Detection of vessel bifurcations in CT scans for automatic objective assessment of deformable image registration accuracy." Medical physics 48.10 (2021): 5935-5946.
- [24]. Wang, Di, et al. "PLOSL: Population learning followed by one shot learning pulmonary image registration using tissue volume preserving and vesselness constraints." Medical image analysis 79 (2022): 102434.
- [25]. Pu J, Gezer N S, Ren S, et al. Automated detection and segmentation of pulmonary embolisms on computed tomography pulmonary angiography (CTPA) using deep learning but without manual outlining[J]. Medical Image Analysis, 2023, 89: 102882.
- [26]. Chu Y, Zhou L, Luo G, et al. Topology-Preserving Computed Tomography Super-Resolution Based on Dual-Stream Diffusion Model[C]//International Conference on Medical Image Computing and Computer-Assisted Intervention. Cham: Springer Nature Switzerland, 2023: 260-270.
- [27]. Bai H, Sadaghianloo N, Gorecka J, et al. Artery to vein configuration of arteriovenous fistula improves hemodynamics to increase maturation and patency[J]. Science translational medicine, 2020, 12(557): eaax7613.
- [28]. Charbonnier J P, Brink M, Ciompi F, et al. Automatic pulmonary artery-vein separation and classification in computed tomography using tree partitioning and peripheral vessel matching[J]. IEEE transactions on medical imaging, 2015, 35(3): 882-892.
- [29]. Nardelli P, Jimenez-Carretero D, Bermejo-Pelaez D, et al. Pulmonary artery–vein classification in CT images using deep learning[J]. IEEE transactions on medical imaging, 2018, 37(11): 2428-2440.
- [30]. Qin Y, Zheng H, Gu Y, et al. Learning tubule-sensitive CNNs for pulmonary airway and artery-vein segmentation in CT[J]. IEEE transactions on medical imaging, 2021, 40(6): 1603-1617.
- [31]. Wittram C. How I do it: CT pulmonary angiography[J]. American journal of roentgenology, 2007, 188(5): 1255-1261.
- [32]. Schoepf U J. Pulmonary artery CTA[J]. Techniques in vascular and interventional radiology, 2006, 9(4): 180-191.
- [33]. Mekada Y, Nakamura S, Ide I, et al. Pulmonary artery and vein classification using spatial arrangement features from X-ray CT images[C]//Proc. 7th Asia-Pacific Conference on Control and Measurement. 2006: 232-235.
- [34]. Zhou, Chuan, et al. "Automatic multiscale enhancement and segmentation of pulmonary vessels in CT pulmonary angiography images for CAD applications." Medical physics 34.12 (2007): 4567-4577.
- [35]. Buelow, Thomas, et al. "Automatic extraction of the pulmonary artery tree from multi-slice CT data." Medical Imaging 2005: Physiology, Function, and Structure from Medical Images. Vol. 5746. SPIE, 2005.
- [36]. Gleeson, T.G., Bulugahapitiya, S., 2004. Contrast-induced nephropathy. American Journal of Roentgenology 183, 1673–1690.
- [37]. Hasebroock, K.M., Serkova, N.J., 2009. Toxicity of mri and ct contrast agents. Expert opinion on drug metabolism & toxicology 5, 403–416.
- [38]. Kabadi A A, Fernandes T M, Papamatheakis D G, et al. Airway hemorrhage complicating pulmonary thromboendarterectomy: risk factors and outcomes[J]. The Annals of Thoracic Surgery, 2023, 116(1): 121-128.
- [39]. Pan L, Li Z, Shen Z, et al. Learning multi-view and centerline topology connectivity information for pulmonary artery–vein separation[J]. Computers in Biology and Medicine, 2023, 155: 106669.
- [40]. Pienn M, Burgard C, Payer C, et al. Healthy lung vessel morphology derived from thoracic computed tomography[J]. Frontiers in physiology, 2018, 9: 346.